

편미분방정식과 응용 (MAT4033)

Lecture notes for Partial Differential Equations and Its Applications (2026 Fall)

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1. Where PDEs Come From

1.1 What is a PDE?

General representation of PDEs is of form

$$F(x, u(x), \partial_{x_i} u(x), \partial_{x_i x_j} u(x), \dots) = 0$$

which is a relation that function $u = u(x)$ depending on n -dimensional variable $x = (x_1, \dots, x_n)$ satisfies. If x is one-dimensional, then it is called an **Ordinary Differential Equation (ODE)**.

Many of the principles or laws underlying the behavior of the nature, such as Newton's law, are statements of relations involving rules at which things happen. In this case, the rules are expressed in terms of derivatives, and the relations are expressed in terms of differential equations.

Examples of PDEs.

1. (Linear Transport Equation)

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = u_t + cu_x = 0 \quad \text{for } c \in \mathbb{R} \text{ and } u = u(x, t).$$

2. (Conservation Law)

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x} [F(u)] = u_t + F(u)_x = 0$$

3. (Diffusion or Heat Equation)

$$\frac{\partial u}{\partial t} - k \frac{\partial^2 u}{\partial x^2} = u_t - ku_{xx} = 0$$

4. (Wave Equation)

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = u_{tt} - c^2 u_{xx} = 0$$

5. (Soliton or KdV Equation)

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = u_t + uu_x + u_{xxx} = 0$$

6. (Linear Schrödinger Equation)

$$\frac{\partial u}{\partial t} + i \frac{\partial^2 u}{\partial x^2} = u_t + iu_{xx} = 0$$

Note that there exist many different types of PDEs. On the above examples, some equations are mentioned to be linear. Which makes a PDE to be linear? Let us define the linearity of a PDE in the following definition.

Definition (Linearity). Let $\mathcal{L}$ be a differential operator.

1. If $\mathcal{L}$ satisfies

$$\mathcal{L}(au + bv) = a\mathcal{L}(u) + b\mathcal{L}(v)$$

for any (smooth) functions u, v and any constants a, b , then $\mathcal{L}$ is called a **linear**.

2. Equations of the form

$$\mathcal{L}u = 0$$

is called a **homogeneous linear equation**.

3. Equations of the form

$$\mathcal{L}u = g \quad \text{with } g \neq 0$$

is called a **inhomogeneous linear equation**.

Examples of Linearity.

1. (Example of Linear Differential Operator)

$$\mathcal{L} = \frac{\partial}{\partial t} + c \frac{\partial}{\partial x}$$

2. (Example of Homogeneous Linear Equation)

$$\mathcal{L} = \frac{\partial}{\partial t} + c \frac{\partial}{\partial x}, \quad \mathcal{L} = \frac{\partial}{\partial t} - k \frac{\partial^2}{\partial x^2}, \quad \mathcal{L} = \frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x^2}$$

Now we talk about the solutions of PDEs. A **solution** of a PDE is a function $u(x_1, \dots, x_n)$ that satisfies the equation identically, at least some region of the $x_1, \dots, x_n$ variables. The following remark is important technique to solve the equation.

Remark.

1. (The Superposition Principle) For $\mathcal{L}u = 0$, if

$$u_1, u_2, \dots, u_n$$

are solutions, then their arbitrary linear combination

$$\sum_{i=1}^n c_i u_i(x)$$

is also a solution.

2. For $\mathcal{L}u = g$, suppose u is a solution and v is the solution to $\mathcal{L}v = 0$. Then $u + v$ is also a solution to $\mathcal{L}u = g$. That is,

$$\text{general solution to } \mathcal{L}u = g \iff \text{particular solution} + \text{general solution to } \mathcal{L}u = 0$$

which we already know from the theory of ODEs.

1.2 First-order Equations

Derivation of linear transport equation

Consider a motion of an object among with constant velocity c . Let $x(t)$ be the position of the object at time t . Then we have

$$\frac{dx}{dt} = c \quad (\text{which is equivalent to } \dot{x} = c)$$

and with simple calculation, we have

$$x(t) = x(0) + ct \quad \text{for some constant } x_0.$$

Note that for this case, we only consider *single* object. What if we have more than one object? Furthermore, if the number of objects is quite large, then we only know about their distribution. This leads us to the following question:

Question. What if we have a distribution $u_0(x)$ of $x(0)$'s? How can we describe the motion of the distribution $u(x, t)$ of $x(t)$'s?

To answer this question, let us derive the linear transport equation. Remember that our goal is to expand the description of the motion from a single object to a distribution. Let $u(\cdot, t)$ be the distribution¹ of $x(t)$. Then, the objects in $[a, b]$ at time t will move to $[a + ch, b + ch]$ after time h . That is,

$$\int_a^b u(x, t) dx = \int_{a+ch}^{b+ch} u(x, t + h) dx \quad \text{for all } a, b \in \mathbb{R} \text{ and } h > 0.$$

The idea of above equality is the number of objects would be conserved. If the distribution of objects moves constantly with c , then the overall distribution of $[a, b]$ will equal the distribution of $[a + ch, b + ch]$ after time h . Now, to see the average density of the objects in $[a, b]$, we divide both sides by $b - a$:

$$\frac{1}{b-a} \int_a^b u(x, t) dx = \frac{1}{b-a} \int_{a+ch}^{b+ch} u(x, t + h) dx$$

Letting $b \rightarrow a$, by FTC²,

$$\begin{aligned} \lim_{b \rightarrow a} \frac{1}{b-a} \int_a^b u(x, t) dx &= \lim_{b \rightarrow a} \frac{F(b) - F(a)}{b-a} = F'(a) = u(a, t) \\ \lim_{b \rightarrow a} \frac{1}{b-a} \int_{a+ch}^{b+ch} u(x, t + h) dx &= u(a + ch, t + h) \end{aligned}$$

we have

$$u(a, t) = u(a + ch, t + h) \quad \text{for all } a \in \mathbb{R} \text{ and } h > 0.$$

Then,

$$\frac{u(a + ch, t + h) - u(a, t)}{h} = 0 \implies \frac{u(a + ch, t + h) - u(a, t + h)}{h} + \frac{u(a, t + h) - u(a, t)}{h} = 0$$

Letting $h \rightarrow 0$, we have³

$$u_t(a, t) + cu_x(a, t) = 0 \quad \text{for all } a \in \mathbb{R}.$$

Here, an object initially at x goes to $x + ct$ after time t . Which is same as an object at position x at time t have moved from $x - ct$ at time 0. Therefore, we have

$$\begin{aligned} u(x, t) = u(x - ct, 0) &\iff u(x + ct, t) = u(x, 0) \\ &\iff u(x(t), t) = u(x, 0) \quad \text{where } \frac{dx}{dt} = c \text{ and } x(0) = x. \end{aligned}$$

¹This means that $u(x, t)$ is the density of objects at position x at time t . That is, the number of objects in $[a, b]$ at time t is given by

$$\int_a^b u(x, t) dx.$$

²We may assume that $F(b) = \int_a^b u(x, t) dx$.

³Indeed, the calculation is as follows:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{u(a + ch, t + h) - u(a, t + h)}{h} &= \lim_{h \rightarrow 0} \frac{u(a + ch, t + h) - u(a, t + h)}{ch} \cdot c = cu_x(a, t) \\ \lim_{h \rightarrow 0} \frac{u(a, t + h) - u(a, t)}{h} &= u_t(a, t) \end{aligned}$$

Method of characteristics

In general, if the velocity is not constant but depends on x and t , then we have

$$u_t + f(x, t)u_x = 0$$

where $f(x, t)$ is the velocity function. Now consider

$$\frac{d}{dt}x(t) = f(x(t), t) \quad \text{with } x(0) = x \implies u(x(t), t) = u(x, 0).$$

Then,

$$\begin{aligned} \frac{d}{dt}[u(x(t), t)] &= (u_t + \frac{d}{dt}x(t)u_x)(x(t), t) \\ &= u_t + f(x, t)u_x \Big|_{(x,t)=(x(t),t)} = 0 \end{aligned}$$

That is, $u(x(t), t) = u(x, 0)$. Hence, if we can write

$$x(0) = g(x(t), t)$$

then

$$u(x, t) = u(g(x, t), 0).$$

Note that

$$u(t) + cu(x) = 0 \quad \text{and} \quad x = x + ct - ct = x(t) - ct.$$

This is called the **method of characteristics** and characteristic equation is given by

$$\frac{dx}{dt} = f(x, t).$$

Each $x(t)$ with $x(0)$ given is called a **characteristic curve**.

Example 1.

$$4u_x - 3u_y = 0 \tag{1}$$

$$u(y, 0) = y^3 \tag{2}$$

Solution. Regard (x, y) as (x, t) , then we have

$$4u_x - 3u_t = 0 \implies u_x - \frac{3}{4}u_t = 0 \tag{3}$$

$$u(t, 0) = t^3 \tag{4}$$

Since the characteristic equation is given by

$$\frac{dt}{dx} = -\frac{3}{4} \implies x(t) = x(0) - \frac{4}{3}t \tag{5}$$

$$\implies x(0) = x(t) + \frac{4}{3}t \tag{6}$$

Therefore, we have

$$u(x, t) = u(x + \frac{4}{3}t, 0) \tag{7}$$

$$= (x + \frac{4}{3}t)^3 \tag{8}$$

□

Example 2.

$$u_x + yu_y = 0 \quad (9)$$

$$u(y, 0) = f(y) \quad (10)$$

Solution. Regard $(x, y) \longleftrightarrow (t, x)$, then we get

$$u_t + xu_x = 0 \quad (11)$$

$$u(x, 0) = f(x). \quad (12)$$

Then characteristic equation is given by

$$\frac{dx}{dt} = x \implies x(t) = x(0)e^t \implies x(0) = x(t)e^{-t} \quad (13)$$

Therefore, we have

$$u(x, t) = u(xe^{-t}, 0) = f(xe^{-t})$$

□

1.3 Other equations**Wave equation**

Consider a flexible, elastic, homogenous string of length l which undergoes relatively small transverse vibrations. Set:

1. $u(x, t)$: displacement from equilibrium at time t and position x .
2. ρ : density of the string, which is assumed to be constant.
3. T : the magnitude of the tension vector.

We may assume:

1. The motion is assumed to be purely transverse.
2. The string is flexible; the tension is directed tangentially along the string.

Consider the part of the string between $x = x_0$ and $x = x_1$. By Newton's law,

$$F = ma$$

we have the longitudinal and transverse components of the force:

$$\begin{aligned} \text{(longitudinal)} \quad & \frac{T}{\sqrt{1 + u_x^2}} \Big|_{x_0}^{x_1} = 0, \\ \text{(transverse)} \quad & \frac{Tu_x}{\sqrt{1 + u_x^2}} \Big|_{x_0}^{x_1} = \int_{x_0}^{x_1} \rho u_{tt} dx. \end{aligned}$$

We also assume that the motion is small;

$$|u_x| \ll 1 \implies \sqrt{1 + u_x^2} \approx 1.$$

Thus,

$$Tu_x \Big|_{x_0}^{x_1} \approx \int_{x_0}^{x_1} \rho u_{tt} dx.$$

and taking the limit as $x_1 \rightarrow x_0$ yields

$$(Tu_x)_x = \rho u_{tt}.$$

If T is constant,

$$u_{tt} = c^2 u_{xx}, \quad c^2 = \sqrt{\frac{T}{\rho}}.$$

where c is called the wave speed.

Diffusion / Heat equation

Imagine a motionless liquid filling a straight pipe and a chemical substance diffusing through the liquid. Let

$u(x, t)$: the concentration of the chemical at time t and position x .

The mass of the chemical in the interval $[x_0, x_1]$ is given by

$$M := \int_{x_0}^{x_1} \rho u \, dx.$$

which implies

$$\begin{aligned} \frac{d}{dt} M &= \int_{x_0}^{x_1} \rho u_t \, dx \\ &= (\text{flow in}) - (\text{flow out}) \text{ at } t \end{aligned}$$

By Fick's law,

$$\begin{aligned} (\text{flow in}) &= -ku(x_0, t) \\ (\text{flow out}) &= -ku(x_1, t) \end{aligned}$$

Taking the limit as $x_1 \rightarrow x_0$ yields

$$u_t = k(u_x)_x = ku_{xx}.$$

1.4 Initial and Boundary Conditions

In general, PDEs can have infinitely many solutions. Hence, to single out one solution, one needs to impose auxiliary conditions which motivated by physical and practical reasons. Such conditions are called **initial condition** and **boundary condition**.

Note that in each physical or real-world problem, there is domain D in which the PDE is valid. Let us consider the following example.

Example (Vibrating String). For the vibrating string,

$$D = [0, l] \implies \partial D = \{0, l\}$$

where ∂D denotes the boundary of D . Needless to say, at the boundary, the string is fixed. Hereby we obtain the following boundary conditions:

$$u(0, t) = c_1 \quad \text{and} \quad u(l, t) = c_2$$

Even if we do not know c_1 and c_2 exactly, we can guess that there must be some constants c_1 and c_2 such that the above boundary conditions hold.

3 Kinds of boundary conditions

There are three kinds of boundary conditions:

1. **Dirichlet condition** : u is specified on the boundary ∂D .
2. **Neumann condition** : Normal derivative is specified as

$$\frac{\partial u}{\partial n} := \nabla u \cdot \mathbf{n}$$

where $\mathbf{n}$ is outward normal vector to the boundary ∂D .

3. **Robin condition :**

$$\frac{\partial u}{\partial n} + \alpha u \text{ is specified}$$

Example of boundary conditions.

1. **(Vibrating String)** As we have seen in the previous example, the boundary conditions are given by

$$u(0, t) = c_1 \quad \text{and} \quad u(l, t) = c_2$$

which is an example of Dirichlet condition.

2. **(Heat Conduction)** The object D is perfectly insulated and the heat is flowing through D . Then, no heat flows across the boundary ∂D . This is an example of Neumann condition since

$$\frac{\partial u}{\partial n} = 0 \quad \text{on } \partial D$$

and it is homogeneous.

Remark. In case, the domain D is unbounded, one might provide continuity conditions at infinity. For example,

$$\lim_{|x| \rightarrow \infty} \rho(x, t) = \rho_\infty \geq 0 \quad \text{and} \quad \int_{\mathbb{R}^d} \rho(x, t) dx = 1, \dots$$

1.5 Well-posedness Problems

For a physical or real-world problem, scientists and engineers impose physically realistic conditions. For example, initial and boundary conditions or regularity of solutions. Then the mathematicians need to check:

1. **Existence :** Whether there exists at least a solution satisfying all the conditions.
2. **Uniqueness :** There is at most one solution.
3. **Stability :** Closely related to uniqueness, the unique solution depends in a stable manner on the initial and boundary data of the problem.

If the PDE problem, which composed with the equation and conditions, with given conditions satisfies all the above three conditions, we say it is **well-posed** or **in the sense of Hadamard**. Otherwise, it is **ill-posed**.

Example of well-posedness.

1. Consider

$$u_{tx} + cu_{xx} = 0, \quad \text{and} \quad u(x, 0) = u_0(x)$$

that is, the initial distribution of $u(x, t)$ at time $t = 0$ is specified with $u_0(x)$. Note that

$$u(x, t) = u_0(x - ct) + f(t)$$

is a solution which implies that the solution is not unique.

2. Consider

$$u_{xx} + u_{yy} = 0, \quad \text{where} \quad D = \{-\infty < x < \infty, 0 < y < \infty\}$$

For each $n \in \mathbb{N}$, impose boundary conditions

$$u_n(x, 0) = 0 \quad \text{and} \quad \frac{\partial u_n}{\partial y}(x, 0) = e^{-\sqrt{n}} \sin(nx)$$

then, the solution is given by

$$u_n(x, y) := \frac{1}{n} e^{-\sqrt{n}} \sin(nx) \sinh(ny).$$

The boundary data converges to zero pointwise as $n \rightarrow \infty$, but the solution, if $y \rightarrow 0$ is fixed, u_n does not tend to zero as $n \rightarrow \infty$. Hence, the solution is not stable.

Chapter 2. Wave and Diffusions

2.1 The wave equation

Consider the following problem:

$$\begin{aligned} u_{tt} - c^2 u_{xx} &= 0, & -\infty < x < \infty, & \quad t > 0, \\ u(x, 0) &= \phi(x), & u_t(x, 0) &= \psi(x). \end{aligned}$$

One may factorize the equation as⁴

$$(\partial_t - c\partial_x)(\partial_t + c\partial_x)u = 0 \quad : \text{factorization of the differential operator}$$

General solution

We need to introduce *characteristic coordinates*:

$$\xi = x + ct, \quad \eta = x - ct \implies \partial_x = \partial_\xi + \partial_\eta, \quad \partial_t = c\partial_\xi - c\partial_\eta.$$

Then we get

$$\partial_t - c\partial_x = -2c\partial_\eta, \quad \partial_t + c\partial_x = 2c\partial_\xi$$

so that we can rewrite the wave equation as

$$u_{tt} - c^2 u_{xx} = 0 \iff -\Delta c^2 \partial_\eta \partial_\xi u = 0.$$

Therefore, we have

$$u = f(\xi) + g(\eta) = f(x + ct) + g(x - ct)$$

Solution to the initial value problem

Note that following approach is called Duhamel's principle for linear transport equations. Let u be a solution to the wave equation and set

$$v := (\partial_t + c\partial_x)u.$$

then

$$\begin{aligned} v(x, 0) &= \partial_t u(x, 0) + c\partial_x u(x, 0) \\ &= \psi(x) + c\phi'(x) \end{aligned}$$

we get

$$v_t - cv_x = 0, \quad \therefore \frac{dx}{dt} = -c.$$

⁴In a higher dimension,

$$\begin{aligned} 0 &= u_{tt} - c^2 \Delta u \\ &= (\partial_t - ic\sqrt{-\Delta})(\partial_t + ic\sqrt{-\Delta})u. \end{aligned}$$

Now, method of characteristics imply

$$v(x - ct, t) = v(x, 0) \quad \text{or,} \quad v(x, t) = v(x + ct, 0) = \psi(x + ct) + c\phi'(x + ct).$$

From $u_t + cu_x = v$, consider the characteristic flow

$$\frac{dx(t)}{dt} = c, \quad x(0) = x$$

implies that

$$\begin{aligned} \frac{d}{dt}(u(x(t), t)) &= (u_t + \frac{dx}{dt}u_x)(x(t), t) \\ &= v(x(t), t) \\ \implies u(x(t), t) &= u(x, 0) + \int_0^t v(x(s), s) ds \\ \iff v(x + ct, t) &= \phi(x) + \int_0^t v(x + cs, s) ds \end{aligned}$$

since $x \longleftrightarrow x + ct$. Then we obtain

$$\begin{aligned} u(x, t) &= u(x - ct, 0) + \int_0^t v(x - c(t - s), s) ds \\ &= \phi(x - ct) + \int_0^t v(x - ct + 2cs, 0) ds \\ &= \phi(x - ct) + \int_0^t \psi(x - ct + 2cs) + c\phi'(x - ct + 2cs) ds \\ &= \phi(x - ct) + \int_{x-ct}^{x+ct} \psi(\tau) + c\phi'(\tau) \frac{d\tau}{2c} \quad (*) \\ &= \frac{1}{2} [\phi(x - ct) + \phi(x + ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(s) ds \quad (: \text{ d'Alembert's formula}) \end{aligned}$$

where $(*)$ we used the change of variable $x - ct + 2cs = \tau \implies ds = \frac{d\tau}{2c}$

Remark.

1. The formula yields a classical solution. That is, u satisfies the equation pointwise. If $\phi, \psi \in C^2$, then u is continuously twice differentiable with respect to x and t . Hence,

$$u \in C^2([0, T] \times \mathbb{R}) \quad \text{for any } T > 0.$$

2. (Causality) Fix $a \in \mathbb{R}$, that is,

$$\phi, \psi \neq 0 \text{ on } u(x, t) \ll 1 \text{ and } 0 \text{ otherwise.}$$

Then after time t ,

$$u(x, t) \neq 0 \text{ if } a \in [x - ct, x + ct]$$

Note that

$$a \in [x - ct, x + ct] \iff |x - a| \leq ct \iff x \in [a - ct, a + ct].$$

thus the information of the initial data at $x = a$ affects the solution on $[a - ct, a + ct]$ which we call the **domain of influence** of the point $(0, a)$. Conversely, fix x, t . Then $u(x, t)$ is determined by the initial data on $[x - ct, x + ct]$ which we call the **domain of dependence** of the point (x, t) .

3. Replace ϕ and ψ by $u(x, t')$ and $u_t(x, t')$. Then

$$\begin{aligned} u(x, t) &= \frac{1}{2} [u(x - c(t - t'), t') + u(x + c(t - t'), t')] + \frac{1}{2c} \int_{x-c(t-t')}^{x+c(t-t')} u_t(s, t') ds \\ &= \frac{1}{2} [u(x - c\Delta t, t') + u(x + c\Delta t, t')] + \frac{1}{2c} \int_{x-c\Delta t}^{x+c\Delta t} u_t(s, t') ds \end{aligned}$$

Then $u(x, t)$ is determined by $[x - ct, x + ct]$.

4. (Energy conservation) Let define the kinetic energy by

$$KE = \frac{1}{2} \int_{-\infty}^{\infty} |u_t(x, t)|^2 dx.$$

If u is smooth and decay sufficiently fast as $|x| \rightarrow \infty$, then

$$\begin{aligned} \frac{d}{dt} KE &= \int_{-\infty}^{\infty} u_t u_{tt} dx \\ &= c^2 \int_{-\infty}^{\infty} u_t u_{xx} dx \\ &= -c^2 \int_{-\infty}^{\infty} u_{xt} u_x dx \\ &= -\frac{c^2}{2} \frac{d}{dt} \int_{-\infty}^{\infty} |u_x|^2 dx \\ &\implies \frac{d}{dt} \left(\frac{1}{2} \int_{-\infty}^{\infty} |u_t|^2 + c^2 |u_x|^2 dx \right) = 0. \end{aligned}$$

Then, the total energy is defined by

$$\frac{1}{2} \int_{-\infty}^{\infty} |u_t|^2 + c^2 |u_x|^2 dx$$

and we can conclude that

$$\frac{d}{dt} \left(\frac{1}{2} \int_{-\infty}^{\infty} |u_t|^2 + c^2 |u_x|^2 dx \right) = 0.$$

Therefore, the total energy is constant and independent to time t .

2.3 The Diffusion Equation

Consider the following diffusion equation

$$u_t = k u_{xx}, \quad 0 \leq x \leq l, \quad t > 0.$$

Then the following theorem holds.

Theorem (Weak Maximum Principle). *If a continuous function $u(s, t)$ ⁵ satisfies the diffusion equation in a rectangle defined by*

$$R := \{(x, t) : 0 \leq x \leq l, 0 \leq t \leq T\},$$

then the maximum value of $u(s, t)$ in the rectangle is assumed either initially ($t = 0$) or on the lateral boundary ($x = 0$ or $x = l$).

⁵Note that $u \in C^1$ in t and C^2 in x is sufficient for the theorem.

Proof. Let M be the maximum value of $u(x, t)$ on three sides

$$t = 0, \quad x = 0, \quad x = l.$$

Claim : $u(x, t) \leq M$ for all $(x, t) \in [0, l] \times [0, T]$

Let $\epsilon > 0$ be an arbitrary positive constant and set

$$v(x, t) := u(x, t) + \epsilon^2 t.$$

Then,

$$\text{WTS : } v(x, t) \leq M + \epsilon l^2 \text{ on } R.$$

Indeed, if so,

$$u(x, t) \leq M + \epsilon^2(l^2 - x^2) \text{ on } R \quad \forall \epsilon > 0 \implies u(x, t) \leq M \text{ on } R.$$

By the definition of $v(x, t)$, we have

$$v(x, t) \leq M + \epsilon l^2 \text{ on } t = 0, x = 0, x = l.$$

Inside the rectangle,

$$\begin{aligned} v_t - kv_{xx} &= u_t - k(u + \epsilon x^2)_{xx} \\ &= u_t - ku_{xx} - 2\epsilon k \\ &= -2\epsilon k < 0. \end{aligned}$$

Now, assume $v(x, t)$ attains its maximum inside the rectangle, which means that there exists

$$(x_0, t_0) \in (0, l) \times (0, T) \text{ such that } v(x, t) \leq v(x_0, t_0) \text{ for all } (x, t) \in R.$$

However, from the calculus, one knows that

$$v_t(x_0, t_0) = 0, \quad v_x(x_0, t_0) = 0, \quad \text{and} \quad v_{xx}(x_0, t_0) \leq 0.$$

Then,

$$(v_t - kv_{xx})(x_0, t_0) \geq 0,$$

which contradicts the fact that $(v_t - kv_{xx})(x_0, t_0) < 0$.

If $v(x, t)$ attains its maximum on the top edge,

$$0 \geq x_0 \geq l, \quad t_0 = T,$$

still we have

$$v_x(x_0, t_0) = 0, \quad v_{xx}(x_0, t_0) \leq 0.$$

Furthermore,

$$v(x_0, t_0) \geq v(x_0, t_0 - \delta) \quad \text{for all } \delta \text{ with } 0 < \delta \leq t_0.$$

This implies that

$$v_t(x_0, t_0) = \lim_{\delta \rightarrow 0^+} \frac{v(x_0, t_0) - v(x_0, t_0 - \delta)}{\delta} \geq 0,$$

we have

$$(v_t - kv_{xx})(x_0, t_0) \geq 0,$$

and which is contradiction again. Therefore, $v(x, t)$ attains its maximum on the lateral boundary or initially. Hence, we have

$$v \leq M + \epsilon l^2 \text{ on } R.$$

□

Remark.

1. In general dimension, let Ω be bounded, open, and connected. We write

$$\Omega_T := \Omega \times (0, T], \quad \Gamma_T := \overline{\Omega_T} \setminus \Omega_T.$$

which denotes Ω the parabolic cylinder and Γ_T the parabolic boundary.

Then, if $u \in C_1^2(\Omega_T) \cap C(\overline{\Omega_T})$ satisfies

$$u_t = k\Delta u \text{ in } \Omega_T,$$

then

$$\max_{\overline{\Omega_T}} u = \max_{\Gamma_T} u.$$

2. (Strong Maximum Principle) If $u(x, t)$ attains a maximum at $(x_0, t_0) \in \Omega_T$, then u is constant in Ω_{t_0} . That is, it is constant up to time t_0 .

Weak maximum principle gives us the uniqueness of the solution of the diffusion equation. We will prove it on the next corollary. Note that it does not guarantee the existence of the solution.

Corollary (Uniqueness of the solution). *Consider the following diffusion equation*

$$\begin{aligned} u_t - ku_{xx} &= f(x, t) \\ u(x, 0) &= \phi(x) \\ u(0, t) &= g(t), \quad u(l, t) = h(t), \quad g, h \in C^1, \quad \phi \in C^2 \end{aligned}$$

If $u_1, u_2 \in C_1^2(\Omega_T) \cap C(\overline{\Omega_T})$ are two solutions to above IBVP, then

$$u_1 \equiv u_2 \quad \text{on } \overline{\Omega_T}.$$

Proof. Let $w := u_1 - u_2$. Then,

$$w_t - kw_{xx} \equiv 0, \quad w(x, 0) = 0, \quad w(0, t) = 0, \quad w(l, t) = 0.$$

By the weak maximum principle, we have

$$w \leq 0 \text{ on } \overline{\Omega_T}.$$

Apply the same argument to $-w$, we are done. □

Remark.

1. (Proof by energy method) We can prove the uniqueness of the solution by energy method. Let $w := u_1 - u_2$, consider

$$\begin{aligned} \frac{1}{2} \frac{d}{dt} \int_0^l w^2 dx &= k \int_0^l ww_{xx} dx \\ &= kw w_x \Big|_{x=0}^{x=l} - k \int_0^l w_x^2 dx \leq 0 \end{aligned}$$

then

$$\int_0^l w^2 dx \leq \int_0^l w^2(x, 0) dx. \tag{*}$$

Thus, if $w(x, 0) \equiv 0$, then

$$\int_0^l w(x, t)^2 dx = 0 \quad \text{for all } t \in [0, T] \text{ and } w \in C(\overline{\Omega_T})$$

which implies that $w \equiv 0$ on $\overline{\Omega_T}$. Furthermore, note that (*) shows that the L^2 -stability estimates with respect to the initial data.

2. (Minimum principle) Under the same assumptions, if u has positive minimum on $\overline{\Omega_T}$, then

$$\min_{\overline{\Omega_T}} u = \min_{\Gamma_T} u.$$

which can be proved by applying the weak maximum principle to $\frac{1}{u}$.

2.4 Diffusion on the whole line

Consider the following diffusion equation

$$\begin{aligned} u_t &= ku_{xx}, \quad (x \in \mathbb{R}, t > 0) \\ u(x, 0) &= \phi(x). \end{aligned}$$

To construct a solution, one needs to investigate its *invariance properties*.

- (a) A translation $u_y(x, t) := u(x - y, t)$ of any solution $u(x, t)$ is another solution.⁶ for any fixed $y \in \mathbb{R}$.
- (b) Any derivative of a solution is again a solution.
- (c) A linear combination of solutions is again a solution.
- (d) An integral of a solution⁷, is again a solution.
- (e) If $u(x, t)$ is a solution, so is the dilated function $u(\sqrt{a}x, at)$ for any $a > 0$.

Remark. 1. (a) + (d) : If $S(x, t)$ solves the equation, so does $S(x - y, t)$ and

$$v(x, t) := \int_{-\infty}^{\infty} S(x - y, t) g(y) dy$$

as long as the integral is justified.

2. Note that for (e), we seek a special solution $Q(x, t)$ which is dilation-invariant. Choose an initial data for Q as

$$Q = \begin{cases} Q(x, 0) = 0, & x < 0 \\ Q(x, 0) = 1, & x > 0 \end{cases}$$

Step 1. We even look for $Q(x, t)$ in the special form

$$Q(x, t) = g(p), \quad p = \frac{x}{\sqrt{4kt}}. \quad (*)$$

so that we can obtain a simpler formula for Q .

Step 2. Put $(*)$ into the diffusion equation. Then we have

$$\begin{aligned} Q_t &= \frac{dg}{dp} \cdot \frac{dp}{dt} = g'(p) \cdot p_t = g'(p) \cdot \left(-\frac{x}{2\sqrt{4kt^3}} \right) = -\frac{p}{2t} g'(p), \\ Q_{xx} &= \frac{dg}{dp} \cdot \frac{dp}{dx} = g'(p) \cdot p_x = g'(p) \cdot \frac{1}{\sqrt{4kt}} = g''(p) \cdot p_x^2 = g''(p) \cdot \frac{1}{4kt}. \end{aligned}$$

Thus, the diffusion equation becomes

$$0 = Q_t - kQ_{xx} = \frac{1}{t} \left(-\frac{p}{2} g'(p) - \frac{1}{4} g''(p) \right)$$

⁶We disregard the initial condition for now.

⁷For example, $\int_0^x u(y, t) dy$

which is equivalent to

$$\frac{1}{4}g''(p) + \frac{1}{2}pg'(p) = 0 \iff g''(p) + 2pg'(p) = 0 \iff (e^{p^2}g'(p))' = 0.$$

Hence, we have

$$g(p) = c_1 \int_0^p e^{-s^2} ds + c_2.$$

So we set

$$Q(x, t) = c_1 \int_0^{\frac{x}{\sqrt{4kt}}} e^{-p^2} dp + c_2.$$

Step 3. Due to the initial condition, if $x > 0$,

$$1 = \lim_{t \searrow 0} Q(x, t) = \lim_{t \searrow 0} \left(c_1 \int_0^{\frac{x}{\sqrt{4kt}}} e^{-p^2} dp + c_2 \right) = c_1 \int_0^\infty e^{-p^2} dp + c_2 = c_1 \frac{\sqrt{\pi}}{2} + c_2$$

and if $x < 0$,

$$0 = \lim_{t \searrow 0} Q(x, t) = \lim_{t \searrow 0} \left(c_1 \int_0^{\frac{x}{\sqrt{4kt}}} e^{-p^2} dp + c_2 \right) = c_1 \int_0^{-\infty} e^{-p^2} dp + c_2 = -c_1 \frac{\sqrt{\pi}}{2} + c_2.$$

Thus, we can find that

$$c_1 = \frac{1}{\sqrt{\pi}}, \quad c_2 = \frac{1}{2}.$$

Hence, if we set

$$Q(x, t) := \frac{1}{2} + \frac{1}{\sqrt{\pi}} \int_0^{\frac{x}{\sqrt{4kt}}} e^{-p^2} dp$$

then, we set

$$u(x, t) := \int_{-\infty}^{\infty} S(x - y, t) \phi(y) dy$$

where

$$S(x, t) := Q_x(x, t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}$$

and $\phi(x)$ is the initial data. Since $\frac{\partial Q}{\partial x}$ solves the equation, u satisfies

$$u_t = ku_{xx}.$$

To check $\lim_{t \searrow 0} u(x, t) = \phi(x)$, we assume that

$$\lim_{|y| \rightarrow \infty} \phi(y) = 0, \quad \phi : \text{ is differentiable } \quad \text{and} \quad \int_{-\infty}^{\infty} |\phi'(y)| dy < \infty.$$

Then one can check that

$$\begin{aligned} u(x, t) &= - \int_{-\infty}^{\infty} S(x - y, t) \phi(y) dy \\ &= - \int_{-\infty}^{\infty} \frac{\partial}{\partial y} y \{ Q(x - y, t) \} \phi(y) dy \\ &= - [Q(x - y, t) \phi(y)]_{y=-\infty}^{y=\infty} + \int_{-\infty}^{\infty} Q(x - y, t) \phi'(y) dy \\ &= \int_{-\infty}^{\infty} Q(x - y, t) \phi'(y) dy. \end{aligned}$$

Therefore, we have

$$\begin{aligned}
\lim_{t \searrow 0} u(x, t) &= \lim_{t \searrow 0} \int_{-\infty}^{\infty} Q(x - y, t) \phi'(y) dy \\
&= \int_{-\infty}^{\infty} \lim_{t \searrow 0} Q(x - y, t) \phi'(y) dy \\
&= \int_{-\infty}^x \phi'(y) dy \\
&= \phi(x).
\end{aligned} \tag{**}$$

Note that we used the dominated convergence theorem in (**), and we get the pointwise convergence of the solution to the initial data.

Now we have the following formula:

$$u(x, t) = \int_{-\infty}^{\infty} S(x - y, t) \phi(y) dy = S(\cdot, t) * \phi(x)$$

Such function S is called the source function, Green's function, fundamental solution, or the heat kernel.

Remark.

1. The source function, if regarded as a family of functions parametrized by t ,⁸ forms a family of good kernels as $t \searrow 0$, that is, it satisfies the following properties:

- (i) $\int_{-\infty}^{\infty} S(x, t) dx = 1$ for all $t > 0$.
- (ii) $\sup_{t > 0} \int_{-\infty}^{\infty} |S(x, t)| dx < \infty$.
- (iii) $\int_{|x| > \delta} S(x, t) dx \rightarrow 0$ as $t \searrow 0$ for each $\delta > 0$.

Moreover,

$$\int_{-\infty}^{\infty} |\phi(x)| dx < \infty \implies \int_{-\infty}^{\infty} |S(\cdot, t) * \phi - \phi| dx \rightarrow 0 \text{ as } t \searrow 0$$

which means the L^1 convergence of the solution formula to the initial data.

2. As $t \rightarrow \infty$,

$$S(x, t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}} \rightarrow 0 \quad \text{uniformly in } x \in \mathbb{R}.$$

since $|S(x, t)| \leq \frac{1}{\sqrt{4\pi kt}}$. Thus, once ϕ is merely integrable, then

$$u := S(\cdot, t) * \phi$$

satisfies the heat equation on $(-\infty, \infty) \times (0, \infty)$ and becomes a smooth function.

3. Reflections and Sources

Diffusion on the half line (Sec. 3.1 and 3.3)

Now we consider the homogeneous diffusion equation with homogeneous Dirichlet boundary condition on the half line:

$$\begin{aligned}
v_t &= kv_{xx}, & (x > 0, t > 0) \\
v(x, 0) &= \phi(x), & (x > 0) \\
v(0, t) &= 0, & (t > 0)
\end{aligned}$$

⁸Not as $S(x, t)$, but as $\{S(\cdot, t)\}_{t > 0}$.

Our strategy is to extend our problem to the whole real line and restrict the solution on $\mathbb{R}$ to $[0, \infty)$. For this, we consider the odd extension ϕ_{odd} of the initial data ϕ ;

$$\phi_{\text{odd}}(x) := \begin{cases} \phi(x), & x > 0 \\ 0, & x = 0 \\ -\phi(-x), & x < 0 \end{cases}$$

Let $u(x, t)$ be the solution of

$$\begin{aligned} u_t &= ku_{xx}, \quad (x \in \mathbb{R}, t > 0) \\ u(x, 0) &= \phi_{\text{odd}}(x), \quad (x \in \mathbb{R}) \end{aligned}$$

then we have

$$u(x, t) = S(\cdot, t) * \phi_{\text{odd}}(x) = \int_{-\infty}^{\infty} S(x - y, t) \phi_{\text{odd}}(y) dy$$

solves the equation.

Our choice is to take

$$v(x, t) := u(x, t)|_{x \geq 0}.$$

One needs to check $u(0, t) = 0$; First, u is smooth once ϕ is integrable.⁹ Furthermore,

$$\begin{aligned} u(-x, t) &= \int_{-\infty}^{\infty} S(-x - y, t) \phi_{\text{odd}}(y) dy \\ &= \int_{-\infty}^{\infty} S(x + y, t) \phi_{\text{odd}}(y) dy \\ &= \int_{-\infty}^{\infty} S(x - y, t) \phi_{\text{odd}}(-y) dy \\ &= - \int_{-\infty}^{\infty} S(x - y, t) \phi_{\text{odd}}(y) dy \\ &= -u(x, t) \end{aligned}$$

recall that S is an even function. Thus, $v(0, t) = u(0, t) = 0$ for all $t > 0$.

If ϕ satisfies good properties, then

$$\lim_{t \searrow 0} v(x, t) = \phi(x)$$

and we can see

$$\begin{aligned} v(x, t) &= \int_{-\infty}^{\infty} S(x - y, t) \phi_{\text{odd}}(y) dy \\ &= \int_0^{\infty} S(x - y, t) \phi(y) dy - \int_{-\infty}^0 S(x - y, t) \phi(y) dy \\ &= \int_0^{\infty} (S(x - y, t) - S(x + y, t)) \phi(y) dy. \end{aligned}$$

This kind of method is called the **method of odd extensions** or the **reflection method**.

Appendix

A. Ordinary Differential Equations

A.1 First-order ODEs.

⁹This is basic assumption for initial data of the diffusion equation.